Pietro Freni

Education

2020-2025 **PhD in Pure Mathematics**, *University of Leeds*
Supervisors Vincenzo Mantova and H. Dugald Macpherson
Thesis Title Structural Investigations in some Classes of o-minimal Fields. https://etheses.whiterose.ac.uk/id/eprint/36168

2017-2019 **Laurea Magistrale in Matematica (M.S.)**, *Università di Pisa*, 110/110 cum laude
Advisor Alessandro Berarducci
Thesis Title Surreal Numbers, Transseries and Omega Maps.

2012-2016 **Laurea Triennale in Matematica (B.S.)**, *Università di Pisa*, 110/110 cum laude
Advisor Enrico Sbarra
Thesis Title Risoluzioni libere di ideali monomiali.

2007-2012 **Maturità Scientifica**, *Liceo Scientifico Archimede, Messina*, 100/100 L

Professional Experience

2026-present **Post-doc**, *Institute of Mathematics, Czech Academy of Sciences*, Pure Mathematics
2025 **Research Fellow**, *University of Leeds*, Pure Mathematics
2025 **Substitute Teacher**, *IISS Antonio Pesenti, Cascina (PI), Italy*, Mathematics and Physics
2020-2023 **Math Tutor**, *Stem-up*
2020 **Substitute Teacher**, *ITCG Fermi, Pontedera (PI), Italy*, Informatics

Research Interests

Model theory, o-minimality, category theory, mathematical logic.

Publications

2021 Berarducci, A., Freni, P. (2021). *On the value group of the transseries.* Pacific Journal of Mathematics, 312(2), 335-354. https://msp.org/pjm/2021/312-2/p04.xhtml, arxiv version: https://arxiv.org/abs/2101.04002

Preprints

2023 *On Vector Spaces with Formal Infinite Sums.* https://arxiv.org/abs/2303.08000, accepted for pubblication in Applied Categorical structures.

2024 *T-convexity, Weakly Immediate Types and T-λ-Spherical Completions of o-minimal Structures.* https://arxiv.org/abs/2404.07646

2024 *On the structure of T-λ-spherical completions of o-minimal fields.* https://arxiv.org/abs/2407.07442

2024 *T-convexly valued o-minimal fields are definably spherically complete.* https://arxiv.org/abs/2411.16706

2025 (j.w. Angus Matthews) *Residually Constructible Extensions.* https://arxiv.org/abs/2501.10033

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2025 *Absorbed Types and Derivations in Exponential o-Minimal Theories.* `https://arxiv.org/abs/2511.13447`

Posters

2022 On formal infinite linear combinations (Presented in different versions at: Model Theory and Applications, Cetraro 2022; Unimod 2022, Leeds)

Talks

2025 *Cuts and Derivations in Exponential o-minimal Fields*. Leeds, Models Seminar, and Imperial College London, SEEMOD2025; *Expansions of Hahn fields by Algebras of generalized power series*. Napoli, INdAM intensive period on Model theory and tame expansions of topological fields, and Bologna, Logic, Categories, and Applications Seminar; *Strong vector spaces and some torsion theories of* $\mathrm{Ind}(\mathrm{Vect}^{\mathrm{op}})$. Bologna, Logic, Categories, and Applications Seminar; *Elementary (bounded) spherical completions of exponential o-minimal fields, and their structure.* Pisa, Logic seminar.

2024 *Weakly immediate types and spherical completions of o-minimal structures.* Burnley, 30th LYMoTS meeting; *T-convexity and T-λ-spherical completions of o-minimal structures.* Manchester Logic Seminar.

2023 *What Should Strong Vector Spaces be?* Online, ItaCa Fest May 2024, Leeds, Impromptu Leeds-MacMaster Model Theory days; *Summability in Algebras of Generalized Power Series.* Leeds, Postgraduate Logic Seminar; *How do you solve a system of polynomial equations?* Leeds, Postgraduate Pure Math Seminar.

2022 *What are Transseries?* Leeds, Postgraduate Pure Math Seminar; *Strong Vector Spaces* Minitalk at LYMOTS/SEEMOD meeting.

2021 *Around exponentiation in fields of generalized power series.* Leeds, Postgraduate Logic Seminar; *Gabrielov's theorem and the model completeness of reals with restricted analytic functions.* Leeds, Postgraduate Pure Math Seminar.

Teaching Experience

2023 Tutoring: Sets, Sequences and Series.

2022 Tutoring: Sets, Sequences and Series. Workshop: Further Linear Algebra.

2021 Tutoring: Numbers and Systems.

Further Experience

2025 Orginiser of the Models Seminar at the University of Leeds.

2023 Organiser of the British Postgraduate Model Theory Conference 2023.

2022-2023 Organiser of the Leeds Postgraduate Logic Seminar.

2022 Helping in the organisation of the Unimod 2022 programme.

Fundings and Awards

2020-2024 School of Mathematics Full-Time Doctoral Scholarship at the School of Mathematics, University of Leeds.

2010-2012 Bronze Medal at the Italian Mathematics Olympiads in years 2010, 2011, 2012.

Computer skills

Coding C, C#, octave, SQL, Lean 4 (basic experience with C++, Java, javascript, M2, AGDA).

Other LaTeX, html, css, Excel.

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Languages

Fluent	Italian (native speaker), English.
Good	German.